

A Shock Train in a Supersonic Duct with a Fixed-Pressure Outlet, Reproduced with PINNeAPple

Viscous compressible finite-volume simulations (HLLC/MUSCL) of Mach-2 duct flow under varying back pressure

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PINNeAPple Labs — Physics AI research notes

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Abstract

A commercial-code visualisation (CONVERGE) of Mach-2 flow in a channel with a fixed-pressure outlet shows a shock train: a sequence of shocks through which the core decelerates while interacting with the wall boundary layers. We reproduce this configuration with a new compressible finite-volume solver added to PINNeAPple (HLLC fluxes, MUSCL reconstruction, viscous terms, SSP-RK2). The solver reproduces the Rankine–Hugoniot jump of a standing normal shock at $M = 2$ (downstream Mach 0.5774 vs 0.5774; pressure ratio 4.500 vs 4.5). In a 2-D duct at $Re = 2 \times 10^4$, the lowest back pressure ($p_b/p_{in} = 2.00$) produces no shock train; in the steady cases the leading edge of the train (first centre-line sonic point) moves upstream from $x/H = 3.51$ to 0.49 as p_b/p_{in} rises from 3.00 to 4.00; at 4.25 the train reaches the inlet (unstart); at 2.50 it had not settled by the end of the run. The trend is robust but the absolute position is not grid-converged: a three-grid study at $p_b/p_{in} = 3$ gives $x/H = 6.97, 4.55, 3.30$ (coarse to fine), a grid-convergence index of 50 % and an extrapolated $x/H \approx 2.0$. In a 3-D square duct at $Re = 10^4$, $p_b/p_{in} = 2.50$ puts the leading edge at $x/H = 1.38$; $p_b/p_{in} = 3.00$ puts the leading edge at $x/H = 0.75$ (four walls add friction and blockage, so a given back pressure pushes the train further upstream than in 2-D).

Keywords: shock train; pseudo-shock; supersonic duct; back pressure; HLLC; finite volume; PINNeAPple

What this study shows — and what it does not

- Shows: a verified solver reproducing the qualitative structure of the target picture and the dependence of the shock-train position on back pressure.
- Does not reproduce the CONVERGE case quantitatively: its geometry, Reynolds number, turbulence model and mesh are not known to us; our flow is laminar at $Re = 2 \times 10^4$, where the boundary layers are thicker than in a turbulent high-Re duct.
- The first shock-train position metric we used was wrong (see Methods); it is kept in the tables so the correction is visible.

1 Background

In an inviscid constant-area duct a normal shock at $M = 2$ raises the pressure by 4.5, and a steady shock can only sit in the duct at exactly that back pressure. Viscosity changes this: the boundary layer separates under the adverse pressure gradient, the single shock becomes a train of weaker, bifurcated shocks (a pseudo-shock), and the compression is spread over several duct heights. The train then settles at a position set by the back pressure, and moves upstream as it rises until it reaches the inlet (unstart) [1–3].

2 Methods

- Solver (new in PINNeAPple: `pinneapple_simulation.numerical_solvers.compressible_fv`): conservative Navier–Stokes, $\gamma = 1.4$, $Pr = 0.72$, power-law viscosity ($\omega = 0.76$), HLLC Riemann solver, MUSCL with minmod limiter, SSP-RK2, CFL 0.4. Boundary conditions: supersonic inflow, no-slip adiabatic walls, and a pressure outlet that fixes p where the outflow is subsonic.
- Verification: Sod shock tube (first-order L1 convergence at the discontinuities) and a standing normal shock at $M = 2$ initialised at the exact jump; after $t = 30$ the shock stayed at $x = 2.00$ (initial 2.0).

- Duct: $L/H = 12$, $M_{in} = 2$, $Re = 2 \times 10^4$ (based on H and inflow), 480×40 cells, run to $t = 45$ flow-through units H/u_{in} ; float32 on Apple MPS.
- Leading-edge position. First attempt: first x where the centre-line pressure exceeds $1.15 p_{in}$. It fired at $x \approx 2.5$ in the $p_b/p_{in} = 2$ case whose core stays supersonic to the exit, because the growing boundary layers compress the core gradually; it therefore does not locate a shock. Used instead (fixed before the other cases were examined): (i) the first centre-line point with $M < 1$, and (ii) the first point where the wall pressure exceeds 1.10 times that of the no-train case on the same grid. A train whose core stays supersonic (oblique train) is detected by (ii) but not (i).
- Grid study: three grids with refinement ratio 1.5 at $p_b/p_{in} = 3.0$; observed order, Richardson extrapolation and GCI from PINNeAPple (pinneapple_analysis.verification.convergence).

3 Results

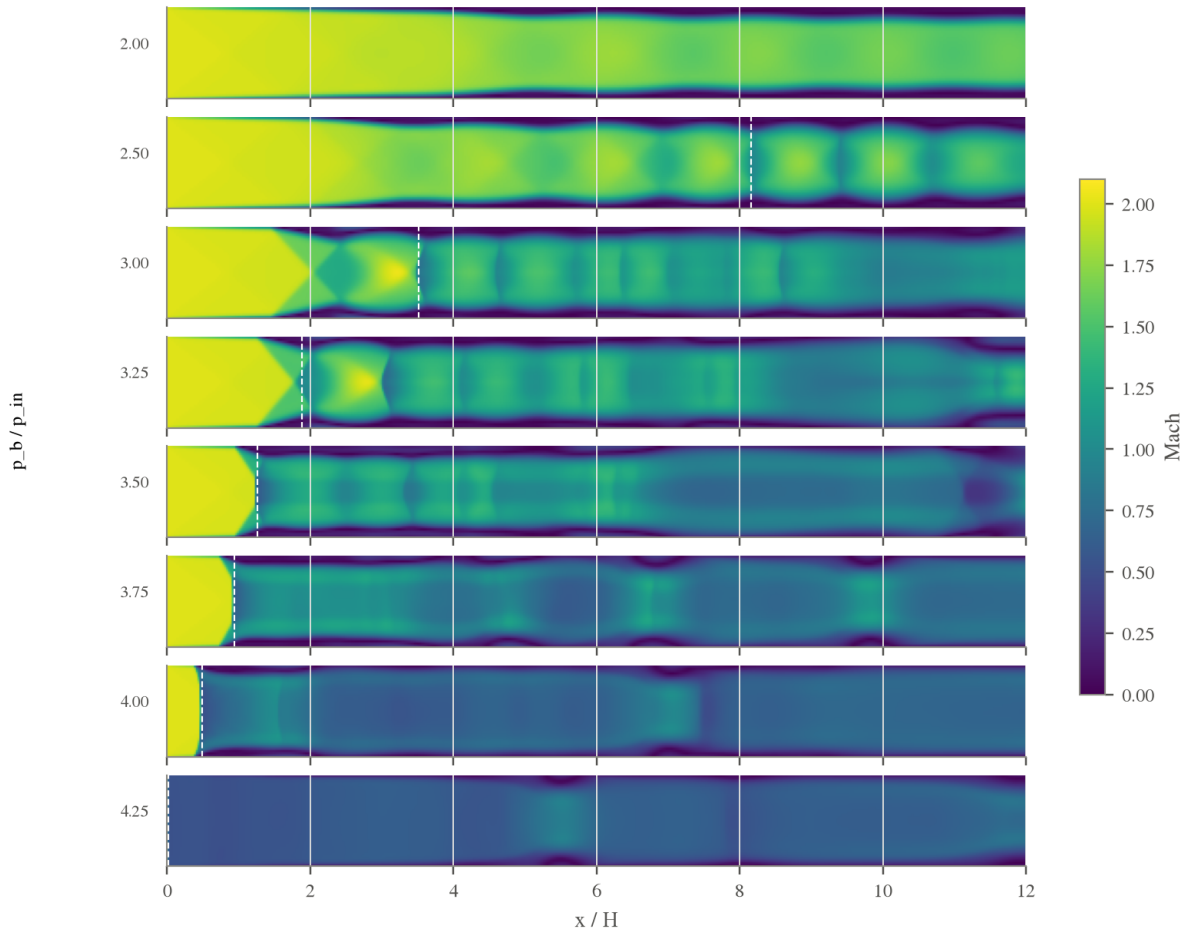


Figure 1. Mach number for each back-pressure ratio (label on the left). Dashed line: centre-line sonic point.

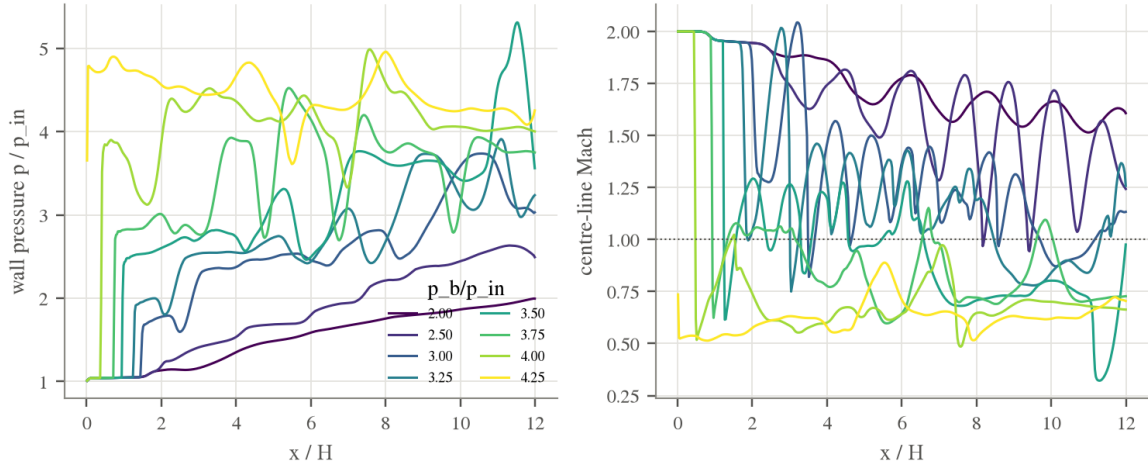


Figure 2. Left: wall pressure. Right: centre-line Mach number (dotted: $M = 1$).

Table 1. Back-pressure sweep. "last-10 range": spread of the sonic-front position over the last 5 time units; above 0.5 H the case is marked unsteady. Wall time depends on the load of the shared machine at the time (up to $10\times$ between identical-cost cases).

p_b / p_{in}	sonic front x/H	last-10 range	wall-rise x/H	first attempt x/H	state	wall time (min)
2.00	—	—	—	2.487	no train	61
2.50	8.162	2.850	2.337	2.487	unsteady	580
3.00	3.513	0.250	1.462	1.962	train	32
3.25	1.887	0.075	1.262	1.712	train	124
3.50	1.262	0.050	0.963	1.238	train	222
3.75	0.938	0.050	0.738	0.912	train	255
4.00	0.487	0.150	0.388	0.463	train	20
4.25	0.013	0.125	0.312	0.013	unstart	19

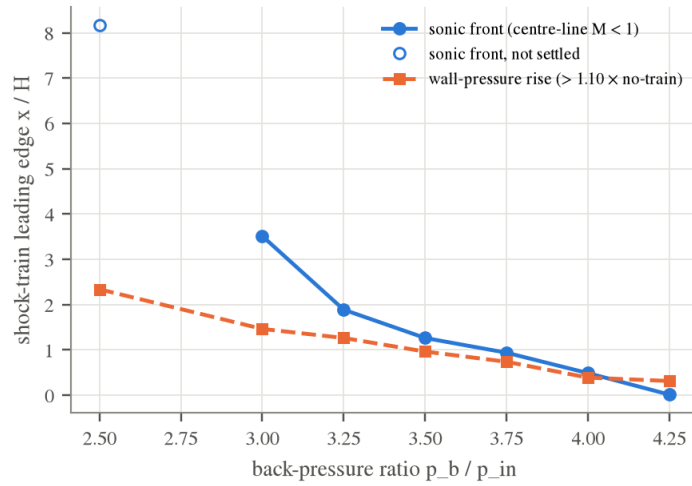


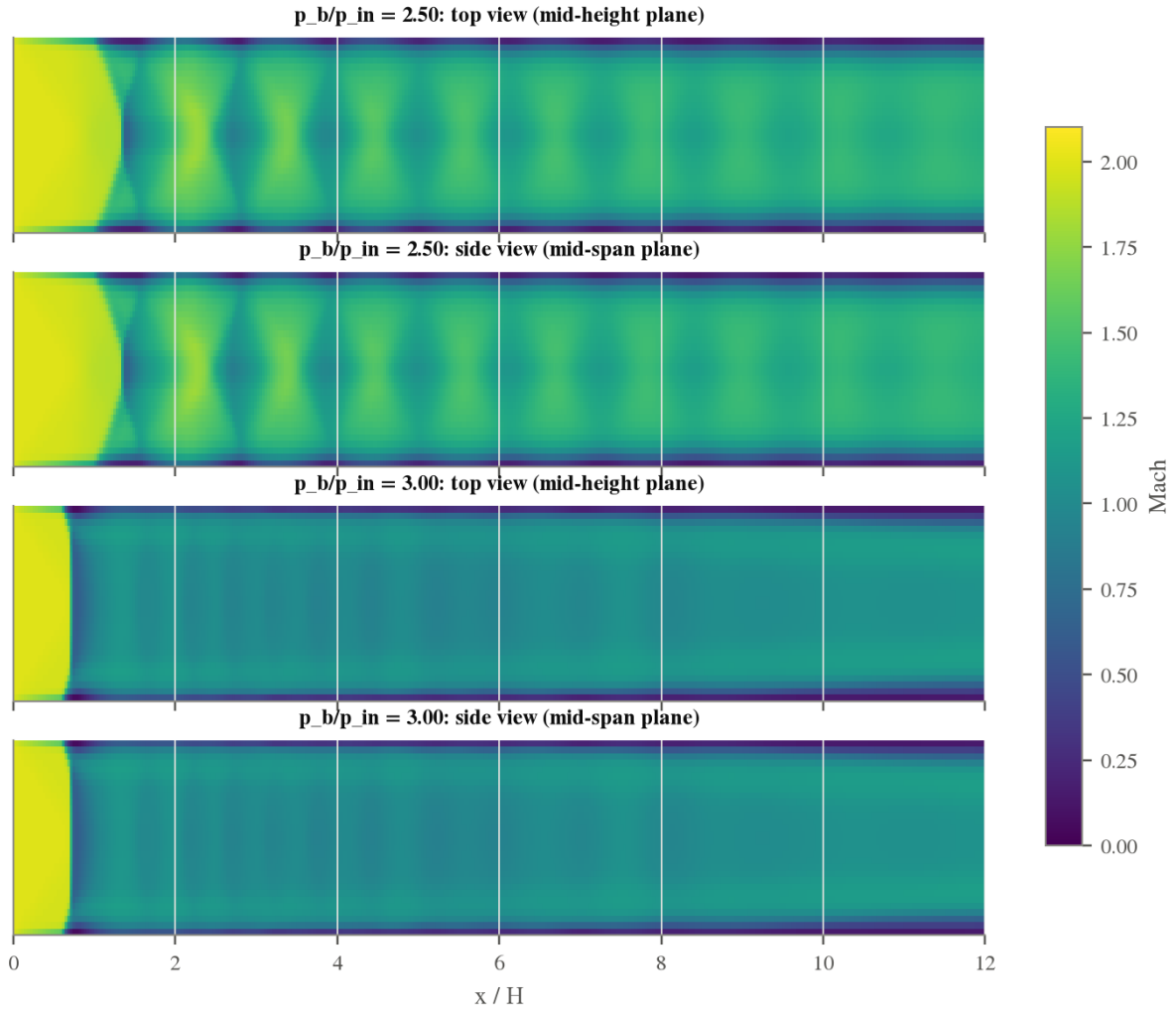
Figure 3. Shock-train leading edge versus back pressure, with both definitions.

Table 2. Grid study at $p_b/p_{in} = 3.0$.

grid	sonic front x/H	first attempt x/H
240×20	6.975	5.475
360×30	4.550	2.950
540×45	3.300	1.744
observed order	1.634	
extrapolated x/H	1.970	
GCI (fine grid)	50.4 %	
in asymptotic range	no	

The three grids are not in the asymptotic range, so the GCI is an indication of grid sensitivity rather than a calibrated error bar.

The leading-edge position moves by 3.7 duct heights between the coarsest and finest grid. The train starts where the laminar wall boundary layer separates, and the separation point depends on how well the thin layer is resolved (20 to 45 cells across the full height here). The sweep ran at an intermediate grid, so its absolute positions carry an uncertainty of the order of one duct height; the ordering of the cases, the no-train and unstart limits, and the upstream movement with back pressure are the results that do not depend on the grid in this study.

**Figure 4.** 3-D square duct, $Re = 10^4$, $360 \times 30 \times 30$ cells: top and side views, the two views of the target picture.

At $p_b/p_{in} = 3.0$ the 3-D train sits almost at the inlet ($x/H = 0.75$), so that case shows a nearly unstarted duct with subsonic flow downstream, not the developed train of the target picture. The four walls add friction and

blockage, and at $Re = 10^4$ the boundary layers are thicker than in the 2-D sweep, so the same back pressure acts more strongly. At $p_b/p_{in} = 2.50$ the leading edge is at $x/H = 1.38$, a train inside the duct as in the target picture.

4 Discussion

The simulations reproduce what the picture shows: a supersonic core that decelerates through a series of shocks, with separated wall regions, whose position is controlled by the fixed outlet pressure. The metric correction is the main methodological lesson. A pressure threshold measured against the inflow value mistakes viscous compression for a shock; any automated reading of shock position from CFD (or from a surrogate trained on it) should reference the no-shock flow or a sonic condition. A surrogate of the leading-edge position over back pressure and Reynolds number was not attempted: eight 2-D cases at one Reynolds number are too few to fit and validate one honestly.

References

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